



COURSE SYLLABUS

CS 305 - Social, Ethical, and Legal Implications of Computing

1. GENERAL INFORMATION

Course name: Social, Ethical, and Legal Implications of Computing

Course name (in Vietnamese):

Course ID: CS305

Knowledge block: Computer Science

Number of credits: 4

Credit hours for theory: 40

Credit hours for practice: 30

Credit hours for self-study: 90

Prerequisite: none

Instructors: Nguyen Van Vu

2. COURSE DESCRIPTION

The advancement of information technology has changed the way we live, interact, and work. However, it also results in social, ethical, and legal issues that we have to address. This course will introduce students important social and professional issues related to information technology and provide opportunities for students to discuss real-world situations where social, ethical, and legal implications are relevant to the information technology profession. The course examines issues of algorithmic bias, privacy, intellectual property, digital rights, and the societal impact of emerging technologies powered by artificial intelligence. Students will engage with real-world case studies, ethical simulations, and debates to develop critical perspectives on responsible technology development.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Recognize social, ethical, and legal issues related to computing.	
G2	Understand major ethical theories and apply them to reason social, ethical, and legal issues related to computing.	
G3	Understand basic legal principles that may be applied to address legal issues.	
G4	Recognize their responsibilities to adhere to the professional and ethical standards.	
G5	Develop argumentation and presentation skills through debates and presentations on topics related to social, ethical, and legal implications of computing.	
G6	Identify and evaluate ethical issues in AI applications, including bias, privacy, and autonomy.	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Students are able to recognize social, ethical, and legal issues related to computer network and network security, intellectual property, privacy, and computer reliability.	T
G2.1	Students understand major ethical theories including Divine Command, Relativism, Utilitarianism, Kantianism, and Social Contract.	T
G2.2	Students can apply the major ethical theories to reason social, ethical, and legal issues related to computing.	U

G3.1	Students can understand basic legal principles that may be applied to address legal issues.	T
G4.1	Students can recognize their responsibilities to adhere to the professional and ethical standards defined by the ACM	T
G4.2	Students can evaluate and decide whether a given action is ethical according to the ethics of the IT professional	U
G5.1	Develop argumentation and presentation skills through debates and presentations on topics related to social, ethical, and legal implications of computing	U
G6.1	Identify and evaluate ethical issues in AI applications, including bias, privacy, and autonomy	U

5. TEACHING PLAN

ID	Topic	Course outcomes	Teaching/Learning Activities	Evaluation (Homework/Task)
1	Class Introduction History of Computing	G1.1	Reading Ch. 1, Appendix B. NSPE Code of Ethics for Engineers ACM Code of Ethics and Professional Conduct	<i>Quizzes are given random in any class.</i> <i>Discussion is held in most class.</i>
2	Ethical Theories	G2.1	Reading Ch. 2 "Social Contract Theory" (Moodle) "Neo-Kantianism" (Moodle) "Act and Rule Utilitarianism" (Moodle)	In-class exercises Lightning presentations

3	Networked Communications	G1.1, G2.2, G3.1	Reading Ch. 3 I Make Video Games. I Won't Let My Daughters Play Them	In-class exercises Lightning presentations
4	Intellectual Property in the Digital Age	G1.1, G2.2, G3.1	Reading Ch. 4 "The GNU Manifesto Turns Thirty" (Moodle)	In-class exercises Lightning presentations
5	Privacy and Surveillance	G1.1, G2.2, G3.1	Reading Ch. 5 and 6 Looking back at the Snowden revelations	In-class exercises Lightning presentations
6	Computer Reliability	G1.1, G2.2, G3.1	Reading Ch. 8 The many human errors that brought down the Boeing 737 Max	In-class exercises Lightning presentations
7	Introduction to AI Ethics	G1.1, G2.2, G3.1	Reading Ch. 10 What is artificial general intelligence (AGI)? Artificial general intelligence: Are we close, and does it even make sense to try? The Invention of 'Ethical AI'	In-class exercises Lightning presentations
8	Algorithmic Bias and Fairness	G6.1	Reading Ch. 8 Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification Algorithmic Decision Making and the Cost of Fairness	In-class exercises Lightning presentations
9	Automation and Globalization	G6.1	Reading Ch. 10	In-class exercises Lightning presentations

10	Emerging Technologies and Ethical Implications	G6.1	Reading Ch. 10 Our Oppenheimer Moment: The Creation of A.I. Weapons	In-class exercises Lightning presentations
11	Students' presentations	G5.1		Presentation due

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Contribution (%)
LT	Lightning presentations (5x10 format)	Every week (starting from Week 2), 5 students will be called randomly to present in 7 minutes + 3 minutes for Q&A. All students need to prepare a topic related to the lectures prior to OR on the presentation week. Students will present individually.	G1.1	20%
DA	Discussions and in-class activities	Participation in class discussion will be graded based on how many discussion sessions attended, how many questions are asked/answered.	G1.1, G2.1, G2.2, G3.1, G4.1, G4.2	20%
Qz	Quizzes	Quizzes will be given randomly during the class. Quizzes consist of short questions on the topics discussed in the current or previous class.	G1.1, G2.1, G2.2, G3.1, G4.1, G4.2	15%
FP	Final presentation	Students will form groups of two students. Each group will deliver a presentation in 15-20 minutes on a topic related to social, professional, and ethical issues.	G5.1	20%

FE	Final Exam	In the final exam, students will answer to exam questions in-person in 60 to 90 minutes.	G6.1	25%
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7. RESOURCES

Required Text

- *Ethics for the Information Age (9th ed.)*, Michael Quinn. Pearson.

Additional Reading/References

- *On Liberty and Utilitarianism*, John Stuart Mill, Bantam Books, NY
- *Philosophy of Science: A Very Short Introduction (2nd ed.)*, Samir Okasha, Oxford University Press
- *Justice: What's the Right Thing to Do* (reprint ed.), Michael Sandel, Farrar, Straus and Giroux
- *Bàn về tự do*, John Stuart Mill, NXB Tri Thuc (Vietnamese)
- *Tự do kinh tế và chính thể đại diện*, F. A. Hayek, NXB Tri Thuc (Vietnamese)
- *Nền dân trị Mỹ*, Alexis De Tocqueville, NXB Tri Thuc (Vietnamese)
- (Other books from the Book Reading Challenge).

Software Tools

- MS Office, Google Docs for preparing presentation, writing documents

8. GENERAL REGULATIONS & POLICIES

- Students are prohibited from plagiarism and cheating of any kind in this class. Some examples of plagiarism and cheating are copying someone else's work, giving your work to others, presenting someone else's ideas, discussing with others during an exam, copying text from sources without proper attribution, and using unauthorized materials. Such violations will result in serious consequences. Those who violates will be graded zero for the course. The incident is then submitted to the school and university for further review.
- All students are responsible for reading and adhering strictly to the regulations and policies of the faculty and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exam.

- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.

Late Submission Policy

- The students are encouraged to submit the assignments on time. But if the deadline is not met, 15% deduction on the assignment total grade is applied for each day late. The students will receive zero if submitting after the third day following the deadline. In exceptional and difficult cases such as illness, the students must obtain the permission for possible extensions from the instructor. In doing so, the students must present reasonable evidence along with the request.

Class Attendance and Material Preparation

- Students are required to attend all classes. According to the APCS program's attendance policy, **you have to attend at least 70% of class sessions to be eligible for taking the final exam.**
- My experience is that those attended classes regularly received higher grades than those who skipped the class often.
- Students are strongly encouraged to ask questions during lecture.
- Students are required to read all reading materials listed in the class schedule, desirably before each classroom lecture.